```
~ $ ssh -i citibike.pem hadoop@3.235.186.53
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@         WARNING: UNPROTECTED PRIVATE KEY FILE!          @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0644 for 'citibike.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "citibike.pem": bad permissions
hadoop@3.235.186.53: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
~ $ chmod 400 citibike.pem
~ $ ssh -i citibike.pem hadoop@3.235.186.53
       #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
Last login: Thu Jan 22 06:38:28 2026

EEEEEEEEEEEEEEEEEEEE MMMMMMMM           MMMMMMMM RRRRRRRRRRRRRRR
E::::::::::::::::::E M::::::M         M::::::M R::::::::::::::R
EE:::::EEEEEEEEE:::E M::::::::M       M::::::::M R:::::RRRRRR:::::R
  E::::E       EEEEE M::::::::::M     M::::::::::M RR::::R      R::::R
  E::::E             M:::::::::::M   M:::::::::::M   R::::R      R::::R
  E::::::EEEEEEEEEE  M:::::M M::::M M::::M M:::::M   R::::RRRRRR:::::R
  E::::::::::::::::E M:::::M  M::::M:::M  M:::::M   R::::::::::::RR
  E::::::EEEEEEEEEE  M:::::M   M:::::M   M:::::M   R::::RRRRRR::::R
  E::::E             M:::::M    M:::M    M:::::M   R::::R      R::::R
  E::::E       EEEEE M:::::M     MMM     M:::::M   R::::R      R::::R
EE:::::EEEEEEEE:::::E M:::::M             M:::::M   R::::R      R::::R
E::::::::::::::::::E M:::::M             M:::::M RR::::R      R::::R
EEEEEEEEEEEEEEEEEEEE MMMMMMM             MMMMMMM RRRRRRR      RRRRRR

[hadoop@ip-172-31-69-51 ~]$ aws s3 ls
2026-01-22 06:18:12 aws-logs-322042279889-us-east-1
2026-01-22 04:20:17 group7-citibike-bucket
```

```
[hadoop@ip-172-31-69-51 ~]$ aws s3 ls s3://group7-citibike-bucket/
                           PRE citibike_parquet/
[hadoop@ip-172-31-69-51 ~]$ df = spark.read.parquet("s3://group7-citibike-bucket/citibike_parquet/")
-bash: syntax error near unexpected token `('
[hadoop@ip-172-31-69-51 ~]$ pyspark
Python 3.9.24 (main, Oct 20 2025, 00:00:00)
[GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] on linux
Type "help", "copyright", "credits" or "license" for more information.
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/01/22 07:03:02 WARN HiveConf: HiveConf of name hive.server2.thrift.url does not exist
26/01/22 07:03:04 WARN Client: Neither spark.yarn.jars nor spark.yarn.archive is set, falling back to uploading libraries under SPARK_HOME.
26/01/22 07:03:23 WARN YarnSchedulerBackend$YarnSchedulerEndpoint: Attempted to request executors before the AM has registered!
Welcome to
```

```
>>> df = spark.read.parquet("s3://group7-citibike-bucket/citibike_parquet/sample")
>>> df.printSchema()
root
 |-- ride_id: string (nullable = true)
 |-- rideable_type: string (nullable = true)
 |-- started_at: string (nullable = true)
 |-- ended_at: string (nullable = true)
 |-- start_station_name: string (nullable = true)
 |-- start_station_id: string (nullable = true)
 |-- end_station_name: string (nullable = true)
 |-- end_station_id: string (nullable = true)
 |-- start_lat: string (nullable = true)
 |-- start_lng: string (nullable = true)
 |-- end_lat: string (nullable = true)
 |-- end_lng: string (nullable = true)
 |-- member_casual: string (nullable = true)
 |-- trip_date: string (nullable = true)
 |-- STATION: string (nullable = true)
 |-- LATITUDE: string (nullable = true)
 |-- LONGITUDE: string (nullable = true)
 |-- ELEVATION: string (nullable = true)
 |-- NAME: string (nullable = true)
 |-- TEMP: string (nullable = true)
 |-- TEMP_ATTRIBUTES: string (nullable = true)
 |-- DEWP: string (nullable = true)
 |-- DEWP_ATTRIBUTES: string (nullable = true)
 |-- SLP: string (nullable = true)
 |-- SLP_ATTRIBUTES: string (nullable = true)
 |-- STP: string (nullable = true)
 |-- STP_ATTRIBUTES: string (nullable = true)
 |-- VISIB: string (nullable = true)
 |-- VISIB_ATTRIBUTES: string (nullable = true)
 |-- WDSP: string (nullable = true)
 |-- WDSP_ATTRIBUTES: string (nullable = true)
 |-- MXSPD: string (nullable = true)
 |-- GUST: string (nullable = true)
 |-- MAX: string (nullable = true)
 |-- MAX_ATTRIBUTES: string (nullable = true)
 |-- MIN: string (nullable = true)
 |-- MIN_ATTRIBUTES: string (nullable = true)
```

```
>>> df.count()
44303209
>>> df.show(5)
+----------------+-------------+-----------------+-----------------+-----------------+-------------------+----------------+-------------------+----------------+-----------------+-----------------+-
```

```
>>> df.filter(col("PRCP") == "999.9").count()
0
>>> df.filter(col("GUST") == "999.9").count()
18546580
>>> df.filter(col("GUST") != "999.9").count()
25756219
>>>
```

```
>>> df_clean = df.filter(
...     col("start_station_id").isNotNull() &
...     (col("start_station_id") != "") &
...     col("end_station_id").isNotNull() &
...     (col("end_station_id") != "")
... )
>>> df_clean.count()
44165579
>>> 44303209-44165579
137630
>>>
```

```
>>> df_clean.count()
26/01/22 10:20:58 WARN SparkStringUtils: Truncated the string representation of a plan since it was too large. This behavior can be adjusted by setting 'spark.sql.debug.maxToStringFields'.
44165579
>>> df.count()
44303209
>>> columns_to_keep = [
...     "ride_id","rideable_type","started_at","ended_at",
...     "start_station_name","start_station_id",
...     "end_station_name","end_station_id",
...     "start_lat","start_lng","end_lat","end_lng",
...     "member_casual","trip_date",
...     "TEMP","DEWP","SLP","STP","VISIB","WDSP",
...     "MXSPD","GUST","MAX","MIN","PRCP","SNDP","FRSHTT"
... ]
>>>
>>> df_clean = df_clean.select(*columns_to_keep)
>>>
```

```
>>> from pyspark.sql.functions import to_timestamp
>>>
>>> df_clean = df_clean \
...     .withColumn("started_at", to_timestamp("started_at")) \
...     .withColumn("ended_at", to_timestamp("ended_at")) \
...     .withColumn("trip_date", col("trip_date").cast("date")) \
...     .withColumn("start_lat", col("start_lat").cast("double")) \
...     .withColumn("start_lng", col("start_lng").cast("double")) \
...     .withColumn("end_lat", col("end_lat").cast("double")) \
...     .withColumn("end_lng", col("end_lng").cast("double")) \
...     .withColumn("TEMP", col("TEMP").cast("double")) \
...     .withColumn("DEWP", col("DEWP").cast("double")) \
...     .withColumn("SLP", col("SLP").cast("double")) \
...     .withColumn("STP", col("STP").cast("double")) \
...     .withColumn("VISIB", col("VISIB").cast("double")) \
...     .withColumn("WDSP", col("WDSP").cast("double")) \
...     .withColumn("MXSPD", col("MXSPD").cast("double")) \
...     .withColumn("GUST", col("GUST").cast("double")) \
...     .withColumn("MAX", col("MAX").cast("double")) \
...     .withColumn("MIN", col("MIN").cast("double")) \
...     .withColumn("PRCP", col("PRCP").cast("double")) \
...     .withColumn("SNDP", col("SNDP").cast("double")) \
...     .withColumn("FRSHTT", col("FRSHTT").cast("int"))
>>> df_clean.printSchema()
```

```
>>> df_clean = df_clean.withColumn(
...     "SNDP",
...     when(col("SNDP") == 999.9, 0).otherwise(col("SNDP"))
... )
>>> df_clean.filter(col("SNDP") == 999.9).count()
0
>>> df_clean = df_clean.withColumn(
...     "GUST",
...     when(col("GUST") == 999.9, 0).otherwise(col("GUST"))
... )
>>> df_clean.filter(col("GUST") == 999.9).count()
0
>>> df_clean.show(5, vertical=True)
```

```
>>> from pyspark.sql.functions import dayofweek
>>>
>>> df_clean = df_clean.withColumn(
...     "weekend",
...     dayofweek("started_at").isin([1,7])
... )
```

```
>>> from pyspark.sql.functions import lit
>>> df_clean = df_clean.withColumn("is_holiday", lit(False))
```

```
>>> df_clean = df_clean.drop("trip_distance")
>>> df_clean.columns
['ride_id', 'rideable_type', 'started_at', 'ended_at', 'start_station_name', 'start_station_id', 'end_station_name', 'end_station_id', 'start_lat', 'start_lng', 'end_lat', 'end_lng', 'member_casual', 'trip_date', 'TE
MP', 'DEWP', 'SLP', 'STP', 'VISIB', 'WDSP', 'MXSPD', 'GUST', 'MAX', 'MIN', 'PRCP', 'SNDP', 'FRSHTT', 'weekend', 'is_holiday']
>>> df_clean.printSchema()
root
 |-- ride_id: string (nullable = true)
 |-- rideable_type: string (nullable = true)
 |-- started_at: timestamp (nullable = true)
 |-- ended_at: timestamp (nullable = true)
 |-- start_station_name: string (nullable = true)
 |-- start_station_id: string (nullable = true)
 |-- end_station_name: string (nullable = true)
 |-- end_station_id: string (nullable = true)
 |-- start_lat: double (nullable = true)
 |-- start_lng: double (nullable = true)
 |-- end_lat: double (nullable = true)
 |-- end_lng: double (nullable = true)
 |-- member_casual: string (nullable = true)
 |-- trip_date: date (nullable = true)
 |-- TEMP: double (nullable = true)
 |-- DEWP: double (nullable = true)
 |-- SLP: double (nullable = true)
 |-- STP: double (nullable = true)
 |-- VISIB: double (nullable = true)
 |-- WDSP: double (nullable = true)
 |-- MXSPD: double (nullable = true)
 |-- GUST: double (nullable = true)
 |-- MAX: double (nullable = true)
 |-- MIN: double (nullable = true)
 |-- PRCP: double (nullable = true)
 |-- SNDP: double (nullable = true)
 |-- FRSHTT: integer (nullable = true)
 |-- weekend: boolean (nullable = true)
 |-- is_holiday: boolean (nullable = false)
```

```
>>> from pyspark.sql.functions import radians, sin, cos, sqrt, atan2, round, col
>>>
>>> R_MILES = 3958.8   # Earth radius in miles
>>>
>>> df_clean = df_clean.withColumn(
...     "trip_distance",
...     round(
...         R_MILES * 2 * atan2(
...             sqrt(
...                 sin((radians(col("end_lat")) - radians(col("start_lat"))) / 2)**2 +
...                 cos(radians(col("start_lat"))) *
...                 cos(radians(col("end_lat"))) *
...                 sin((radians(col("end_lng")) - radians(col("start_lng"))) / 2)**2
...             ),
...             sqrt(
...                 1 - (
...                     sin((radians(col("end_lat")) - radians(col("start_lat"))) / 2)**2 +
...                     cos(radians(col("start_lat"))) *
...                     cos(radians(col("end_lat"))) *
...                     sin((radians(col("end_lng")) - radians(col("start_lng"))) / 2)**2
...                 )
...             )
...         ),
...         2   # round to 2 decimal places
...     )
... )
```

```
>>> holiday_dates = [
...     "2024-01-01",  # New Year's Day
...     "2024-01-15",  # Martin Luther King Jr. Day
...     "2024-02-19",  # Washington's Birthday (Presidents' Day)
...     "2024-05-27",  # Memorial Day
...     "2024-06-19",  # Juneteenth National Independence Day
...     "2024-07-04",  # Independence Day
...     "2024-09-02",  # Labor Day
...     "2024-10-14",  # Columbus Day
...     "2024-11-11",  # Veterans Day
...     "2024-11-28",  # Thanksgiving Day
...     "2024-12-25"   # Christmas Day
... ]
>>> df_clean = df_clean.withColumn(
...     "is_holiday",
...     when(col("trip_date").isin(holiday_dates), True).otherwise(False)
... )
>>> df_clean.groupBy("is_holiday").count().show()
+----------+--------+
|is_holiday|   count|
+----------+--------+
|      true|  967856|
|     false|43197723|
+----------+--------+
```

```
>>> df_clean.groupBy("FRSHTT").count().show()
+------+--------+
|FRSHTT|   count|
+------+--------+
|  NULL|     364|
| 10000|11683328|
| 11000|  318473|
|101000|  113855|
|110000| 2141606|
|111000|   31930|
|     0|29738141|
|  1000|  137882|
+------+--------+

>>> from pyspark.sql.functions import unix_timestamp
>>>
>>> df_clean = df_clean.withColumn(
...     "trip_duration_min",
...     (unix_timestamp("ended_at") - unix_timestamp("started_at")) / 60
... )
>>> df_clean.select("trip_duration_min").show(5)
+------------------+
| trip_duration_min|
+------------------+
|               4.0|
|             16.55|
|              35.8|
|25.166666666666668|
| 44.38333333333333|
+------------------+
only showing top 5 rows
```

```
>>> df_clean = df_clean.withColumn("trip_duration_min",round(col("trip_duration_min"),2))
>>> df_clean.select("trip_duration_min").show(5)
+------------------+
|trip_duration_min|
+------------------+
|               4.0|
|             16.55|
|              35.8|
|             25.17|
|             44.38|
+------------------+
only showing top 5 rows
```

```
>>> df_clean.select("started_at","ended_at","trip_distance","trip_duration_min").filter(col("trip_duration_min") >= 1440).show(20, truncate=False)
+-----------------------+-----------------------+-------------+-----------------+
|started_at             |ended_at               |trip_distance|trip_duration_min|
+-----------------------+-----------------------+-------------+-----------------+
|2024-01-06 14:41:10.083|2024-01-07 14:50:52.68 |2.18         |1449.7           |
|2024-01-07 09:31:20.844|2024-01-08 10:14:51.719|0.67         |1483.52          |
|2024-01-08 07:58:14.295|2024-01-09 08:27:37.544|4.69         |1469.38          |
|2024-01-08 17:39:38.677|2024-01-09 18:30:19.572|0.65         |1490.68          |
|2024-01-08 18:07:49.854|2024-01-09 18:30:19.572|1.03         |1462.5           |
|2024-01-08 18:18:18.791|2024-01-09 18:30:19.572|0.77         |1452.02          |
|2024-01-08 18:17:42.765|2024-01-09 18:30:19.572|2.06         |1452.62          |
|2024-01-08 17:54:15.987|2024-01-09 18:31:30.801|0.43         |1477.25          |
|2024-01-08 18:29:49.172|2024-01-09 18:30:19.572|0.91         |1440.5           |
|2024-01-08 17:47:34.225|2024-01-09 18:30:19.572|1.07         |1482.75          |
|2024-01-08 17:42:46.458|2024-01-09 18:30:19.572|1.31         |1487.55          |
|2024-01-08 18:09:59.653|2024-01-09 18:31:32.997|0.83         |1461.55          |
|2024-01-08 17:46:11.893|2024-01-09 18:30:19.572|0.9          |1484.13          |
|2024-01-08 17:44:33.056|2024-01-09 18:30:19.572|0.58         |1485.77          |
|2024-01-04 08:47:58.297|2024-01-05 09:26:23.665|0.92         |1478.42          |
|2024-01-13 13:14:46.32 |2024-01-14 13:15:09.469|1.5          |1440.38          |
|2024-01-11 09:22:08.383|2024-01-12 09:55:42.204|0.63         |1473.57          |
|2024-01-09 08:38:33.292|2024-01-10 08:42:34.565|0.91         |1444.02          |
|2024-01-08 19:34:07.217|2024-01-09 20:33:13.307|1.08         |1499.1           |
|2024-01-02 08:41:03.805|2024-01-03 09:25:44.636|0.82         |1484.68          |
+-----------------------+-----------------------+-------------+-----------------+
only showing top 20 rows
```

```
>>> out_of_bounds_df = df_clean.filter(
...     ~(
...         (col("start_lat").between(40.0, 41.0)) &
...         (col("end_lat").between(40.0, 41.0)) &
...         (col("start_lng").between(-74.0, -73.4)) &
...         (col("end_lng").between(-74.0, -73.4))
...     )
... )
>>>
```

```
>>> out_of_bounds_df.count()
9029098
>>> df_clean.filter(col("trip_duration_min") >= 1440).count()
380
```

```
>>> df_clean = df_clean.filter(
...     (col("start_lat").between(40.40, 40.95)) &
...     (col("end_lat").between(40.40, 40.95)) &
...     (col("start_lng").between(-74.30, -73.60)) &
...     (col("end_lng").between(-74.30, -73.60))
... )
>>> from pyspark.sql.functions import col, sum, when
>>>
>>> null_counts = df_clean.select([
...     sum(when(col(c).isNull(), 1).otherwise(0)).alias(c)
...     for c in df_clean.columns
... ])
>>>
>>> null_counts.show(truncate=False)
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-
--+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-
|ride_id|rideable_type|started_at|ended_at|start_station_name|start_station_id|end_station_name|end_station_id|start_lat|start_lng|end_lat|end_lng|member_casual|trip_date|TEMP|DEWP|SLP|STP|VISIB|WDSP|MXSPD|GUST|MAX|M
IN|PRCP|SNDP|FRSHTT|weekend|is_holiday|trip_distance|trip_duration_min|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-
--+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-
|0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |0      |
64|364 |364 |364   |0      |0      |0      |0      |0      |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-
--+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-
```

```
>>> out_of_bounds_df = df_clean.filter(
...     ~(
...         (col("start_lat").between(40.40, 40.95)) &
...         (col("end_lat").between(40.40, 40.95)) &
...         (col("start_lng").between(-74.30, -73.60)) &
...         (col("end_lng").between(-74.30, -73.60))
...     )
... )
>>>
>>> out_of_bounds_df.count()
178
>>> out_of_bounds_df.select("start_lat","start_lng","end_lat","end_lng").show(20, truncate=False)
+-----------------+-----------------+-------+-------+
|start_lat        |start_lng        |end_lat|end_lng|
+-----------------+-----------------+-------+-------+
|40.73266         |-73.95826        |0.0    |0.0    |
|40.73564         |-73.95866        |0.0    |0.0    |
|40.70767788      |-73.94016171     |0.0    |0.0    |
|40.69385         |-73.90974        |0.0    |0.0    |
|40.68900443      |-73.96023854     |0.0    |0.0    |
|40.68018         |-73.91136        |0.0    |0.0    |
|40.65272         |-73.94994        |0.0    |0.0    |
|40.7552433       |-73.9485678      |0.0    |0.0    |
|40.697787        |-73.973736       |0.0    |0.0    |
|40.68012         |-73.93168        |0.0    |0.0    |
|40.68851534      |-73.9647628      |0.0    |0.0    |
|40.70539         |-73.92976        |0.0    |0.0    |
|40.7031724       |-73.940636       |0.0    |0.0    |
|40.68012         |-73.93168        |0.0    |0.0    |
|40.67524         |-73.93596        |0.0    |0.0    |
|40.67524         |-73.93596        |0.0    |0.0    |
|40.7652654       |-73.98192338     |0.0    |0.0    |
|40.70962789682351|-73.93145710229872|0.0    |0.0    |
|40.67524         |-73.93596        |0.0    |0.0    |
|40.67524         |-73.93596        |0.0    |0.0    |
+-----------------+-----------------+-------+-------+
only showing top 20 rows
```